Junho Andrew Choi

Berkeley, CA | 510-734-9646 | andrewc27@berkeley.edu
linkedin.com/in/junho-choi-a8b097239 | github.com/andrewcjh27 | andrewcjh27.github.io/Portfolio-site

SUMMARY

Data Science and Integrative Biology senior at UC Berkeley targeting data science and machine learning roles in biotechnology and healthcare. Deep learning on 3D medical imaging with a co-authored paper, sequence modeling on protein data, and bench experience running the assays that generate biological data, so model decisions stay legible to the clinicians and scientists who act on them.

EDUCATION

University of California, Berkeley Expected May 2027
B.A. Data Science (Concentration: Computational Methods in Biology) and B.A. Integrative Biology *Berkeley, CA*
Coursework: Practical Genomics (IB 134L, in progress), Physiology (IB 131, in progress), Machine Learning (CS 189, in progress), Principles & Techniques of Data Science, Probability for Data Science, Data Structures

SKILLS

Machine Learning: Python, PyTorch, scikit-learn, NumPy, pandas, 3D U-Net, ResNet3D, CNN-BiLSTM, vision transformers, model interpretability (SHAP, Grad-CAM), feature engineering, model evaluation
Biomedical Data: medical imaging (3D CT), clinical records, microarray and expression data (GEOparse), protein sequence data, survival modeling, PCA, hypothesis testing, SQL, R
Laboratory: PCR, FACS, HPLC, cell culture, in-vivo testing, immunological assays
Spoken: Korean (native), English (fluent), Mandarin

EXPERIENCE

SmartMind AI May 2026 - Aug 2026
AI Researcher, Data Science Team *Seoul, Republic of Korea*
- Built a multi-trial reproducibility framework for a production large language model system, replacing single-run pass/fail testing with measured evaluation and surfacing 3 systemic defects that produced confident but incorrect output
- Audited knowledge-graph quality in a Neo4j ontology pipeline by tuning entity-merge accuracy and similarity thresholds, tracing failed queries to missing or duplicate nodes and migrating domain rules from prose prompts into structured graph nodes

Asan Medical Center (AMC) May 2025 - Aug 2025
AI Research Intern *Seoul, Republic of Korea*
- Co-authored a research paper on a multimodal deep survival model for hepatocellular carcinoma, developing and fine-tuning 3D U-Net and ResNet3D architectures in PyTorch
- Built automated preprocessing pipelines for 3D CT imaging and clinical records, standardizing ingestion across modalities for the research team
- Ran large-scale exploratory analysis for the VLM3D Multi-Abnormality Classification Challenge, informing the feature selection and strategy that placed 7th among multi-institutional teams, and applied SHAP and Grad-CAM so clinicians could see what the models were keying on

Biosolution Co., Ltd. May 2024 - Aug 2024
Research Intern *Republic of Korea*
- Investigated microRNA regulators of osteoarthritis by designing and running PCR, FACS, and HPLC assays to characterize candidate therapeutic targets at the cellular level
- Performed cell culture and in-vivo testing across the preclinical workflow and supported the CAR T-cell therapy program through immunological assays and process documentation

UCSF Aug 2021 - May 2022
Telehealth Coordinator *San Francisco, CA*
- Built longitudinal analyses of telehealth utilization, surfacing the workflow bottlenecks that informed clinical delivery and resource allocation decisions

PROJECTS

- **Activation Domain Prediction Pipeline:** developed a CNN-BiLSTM model classifying transcriptional activation domains from protein sequences with sliding-window feature engineering, SHAP interpretability, and unsupervised K-means clustering, reaching F1 = 0.95 across 10 distinct subclasses
- **Parkinson Disease Biomarker Analysis:** analyzed the GDS2519 microarray dataset with GEOparse, PCA, and Welch t-tests to identify differentially expressed candidate biomarkers, presented as a research poster
- **rad-DINO:** implemented and ran experiments with a self-supervised vision transformer for radiology imaging, evaluating learned representations for downstream classification in PyTorch

LEADERSHIP & AWARDS

US Army, 8th Army 2ID HHBN Sep 2022 - Mar 2024
Non-Commissioned Officer *Republic of Korea*
- Led a 40+ soldier platoon as a bilingual Korean-English platoon leader to a 96% Army Combat Fitness Test pass rate